

Effat University

College of Engineering

CS3151 - Software Engineering

Final Project Report

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GiftCompass

Intelligent Gift Recommendation System

Guiding you to the right gift

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Abstract

GiftCompass is a smart gift recommendation system designed to simplify the process of choosing suitable gifts for different recipients and occasions. The system collects details such as the recipient's interests, relationship to the user, occasion, age group, and budget. It then applies a recommendation approach to filter and rank suitable gift options from a structured gift catalog. This report presents the complete software engineering process for GiftCompass, including the project objective, sustainability alignment, feasibility study, market analysis, requirements specification, design patterns, use case analysis, system diagrams, implementation plan, machine learning approach, testing strategy, and final outcomes. The goal of GiftCompass is to save users time, reduce uncertainty, and help them make more thoughtful gift choices through a simple and personalized digital platform.

Acknowledgment

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Chapter 1

Introduction

1.1 Project Overview

GiftCompass is a web-based recommendation system that helps users find meaningful and suitable gifts. Choosing a gift can be stressful because users may not know what the recipient likes, what is appropriate for the occasion, or what fits their budget. GiftCompass solves this problem by asking users to enter recipient information and then generating ranked gift suggestions.

The platform supports two main groups of users. General users search for gift ideas and receive recommendations, while administrators manage the gift catalog, categories, and system data. The system is designed to be simple, responsive, and accessible through modern web browsers.

1.2 Problem Statement

Many users spend a long time searching across different websites for gift ideas. Existing platforms such as Amazon, Etsy, and Pinterest provide gift inspiration, but most recommendations are category-based or product-driven rather than truly focused on the recipient's personal information. GiftCompass focuses on recipient-centered gift discovery by using structured inputs such as age, relationship, interests, occasion, and budget.

1.3 Project Objectives

The main objectives of GiftCompass are:

- Provide personalized gift recommendations based on recipient information.
- Reduce the time and effort needed to find suitable gifts.
- Support search and filtering by category, price range, occasion, and interests.
- Allow users to save favorite gift ideas for later review.
- Allow administrators to add, update, and delete gift items.
- Build a scalable system that can later support stronger AI-based recommendation features.

1.4 Project Scope

GiftCompass will be implemented as a cloud-based web platform. The system includes account management, recipient profile management, gift recommendation, catalog browsing, search and filtering, favorites, and admin catalog management. Future versions may integrate e-commerce APIs and advanced machine learning models.

Chapter 2

Sustainability Goals

2.1 SDG Alignment

GiftCompass aligns with several Sustainable Development Goals (SDGs):

- **SDG 9 - Industry, Innovation and Infrastructure:** GiftCompass promotes digital innovation through a recommendation-based web system.
- **SDG 12 - Responsible Consumption and Production:** The system encourages users to choose more suitable gifts, reducing unnecessary or unwanted purchases.
- **SDG 8 - Decent Work and Economic Growth:** GiftCompass can support small businesses and local sellers by recommending diverse gift items in future versions.

2.2 Vision 2030 Contribution

GiftCompass supports Saudi Vision 2030 through digital transformation and entrepreneurship. It encourages the development of user-centered digital services, supports innovation in e-commerce, and can later promote local gift shops and small businesses through catalog integration.

Chapter 3

Feasibility Study

3.1 Market Feasibility

GiftCompass has strong market feasibility because gifting is a common activity across birthdays, graduations, weddings, holidays, and personal celebrations. Users often need quick and thoughtful suggestions. Competitor platforms provide inspiration, but GiftCompass differentiates itself by focusing on structured recipient details and personalized ranking.

3.2 Technical Feasibility

The system can be developed using widely available technologies. The frontend can be built with HTML, CSS, JavaScript, or React. The backend can be implemented using Python Flask/Django or Node.js. A relational database such as MySQL or PostgreSQL can store users, gifts, recipient profiles, favorites, and recommendations.

3.3 Economic Feasibility

GiftCompass can start as a low-cost academic prototype using open-source tools and cloud hosting. The main costs are development time, hosting, database services, design tools, and future API integrations.

Category	Estimated Cost
Development Tools	Free/open-source
Cloud Hosting Prototype	Low cost or student tier
Database	Free tier or local database
UI/UX Design	Free design tools
Future API Integrations	Depends on provider

Table 3.1: Economic Feasibility Summary

3.4 System Requirements

3.4.1 Hardware Requirements

Component	Specification
Development Device	Laptop or desktop with at least 8GB RAM and modern processor
User Device	Desktop, laptop, tablet, or smartphone
Network	Stable internet connection
Server	Cloud or local server for prototype hosting

Table 3.2: Hardware Requirements

3.4.2 Software Requirements

Category	Requirement
Operating System	Windows, macOS, or Linux
Frontend	HTML, CSS, JavaScript, React optional
Backend	Python Flask/Django or Node.js
Database	MySQL, PostgreSQL, or SQLite for prototype
Browser	Chrome, Edge, Safari, or Firefox
Security	HTTPS, password hashing, authentication and authorization

Table 3.3: Software Requirements

Chapter 4

Market Analysis

4.1 Competitor Analysis

Feature	GiftCompass	Amazon Gift Finder	Etsy Gift Guides	Pinterest
Recipient-based input	Yes	Partial	Partial	No
Occasion-based suggestions	Yes	Yes	Yes	Yes
Budget filtering	Yes	Yes	Yes	No
Interest-based ranking	Yes	Partial	Partial	No
Favorites saving	Yes	Yes	Yes	Yes
Admin-managed catalog	Yes	No	No	No
Personalized recommendation focus	Yes	Partial	Partial	No

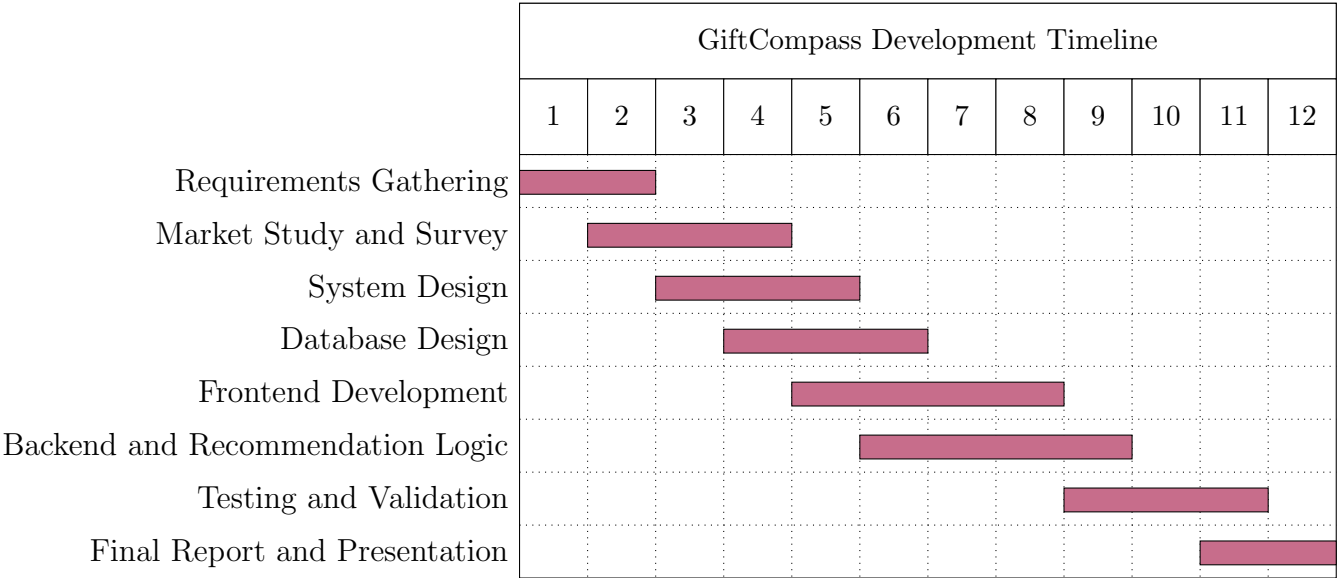
Table 4.1: Competitive Feature Comparison

4.2 Target Users

- Students looking for affordable gifts.
- Busy professionals who need quick recommendations.
- Family members choosing gifts for birthdays and holidays.
- Friends searching for personalized gifts.
- Administrators managing catalog content.

Chapter 5

Project Timeline



Chapter 6

Requirements Specification

6.1 Functional Requirements

ID	Description
RQ-F1	The system shall allow users to register using email and password.
RQ-F2	The system shall allow registered users to log in and log out securely.
RQ-F3	The system shall allow users to create, edit, and delete recipient profiles.
RQ-F4	The system shall generate personalized gift recommendations based on age, interests, relationship, occasion, and budget.
RQ-F5	The system shall allow users to search and filter gifts by category, price range, occasion, and keywords.
RQ-F6	The system shall allow users to save and manage favorite gifts.
RQ-F7	The system shall allow administrators to add, update, delete, and categorize gift items.
RQ-F8	The system shall provide basic analytics about popular gifts and categories for administrators.

Table 6.1: Functional Requirements

6.2 Non-Functional Requirements

ID	Description
RQ-NF1	95% of user requests shall respond within 2 seconds.
RQ-NF2	Gift recommendations shall be generated within 3 seconds after user input.
RQ-NF3	The system shall maintain at least 99.5% monthly up-time.
RQ-NF4	User passwords shall be securely hashed and sensitive data transmitted via HTTPS.
RQ-NF5	The interface shall be responsive on desktop and mobile browsers.
RQ-NF6	The system shall be modular and maintainable for future updates.
RQ-NF7	The system shall support at least 500 concurrent users.

Table 6.2: Non-Functional Requirements

Chapter 7

Design Patterns

7.1 Strategy Pattern

GiftCompass can use the Strategy Pattern to select different recommendation strategies. For example, recommendations may be based on interest matching, budget matching, occasion matching, or a weighted hybrid method. This makes the recommendation engine flexible and easy to extend.

Listing 7.1: Strategy Pattern for Gift Recommendations

```
1 from abc import ABC, abstractmethod
2
3 class RecommendationStrategy(ABC):
4     @abstractmethod
5     def recommend(self, gifts, recipient):
6         pass
7
8 class InterestBasedStrategy(RecommendationStrategy):
9     def recommend(self, gifts, recipient):
10         return [gift for gift in gifts if gift.category in
11                 recipient.interests]
12
13 class BudgetBasedStrategy(RecommendationStrategy):
14     def recommend(self, gifts, recipient):
15         return [gift for gift in gifts if gift.price <= recipient
16                 .budget]
17
18 class RecommendationEngine:
19     def __init__(self, strategy):
20         self.strategy = strategy
21
22     def set_strategy(self, strategy):
23         self.strategy = strategy
24
25     def generate(self, gifts, recipient):
```

```
24         return self.strategy.recommend(gifts, recipient)
```

7.2 Observer Pattern

The Observer Pattern can be used to notify users when new suitable gifts are added, when saved items change price, or when an administrator updates the catalog.

Listing 7.2: Observer Pattern for Gift Alerts

```
1 class UserObserver:
2     def update(self, message):
3         print("Notification:", message)
4
5 class GiftCatalog:
6     def __init__(self):
7         self.observers = []
8
9     def subscribe(self, observer):
10        self.observers.append(observer)
11
12    def notify_all(self, message):
13        for observer in self.observers:
14            observer.update(message)
15
16    def add_gift(self, gift_name):
17        self.notify_all(f"New gift added: {gift_name}")
```

Chapter 8

Use Case Analysis

8.1 Actors

- **User:** Searches for gift ideas, enters recipient details, views recommendations, saves favorites, adds gifts to cart, and completes checkout.
- **Administrator:** Manages the gift catalog, inventory, offers, users, and reports.
- **Recommendation Engine:** Processes user inputs and ranks suitable gifts.
- **Database:** Stores users, recipient profiles, gifts, favorites, inventory, orders, and recommendation results.

8.2 Use Case List

ID	Use Case	Description
UC-01	Get Gift Recommendations	User receives ranked gift suggestions.
UC-02	Search and Filter Gifts	User filters gifts by budget, occasion, category, and interest.
UC-03	Save Favorite Gifts	User saves preferred gifts for later.
UC-04	Manage Gift Catalog	Admin adds, edits, and deletes gift items.
UC-05	View Analytics	Admin views system statistics and popular categories.
UC-06	Create Account	User creates an account using email and password.
UC-07	Login/Logout	User securely enters or exits the system.
UC-08	Manage Profile	User updates personal and recipient information.
UC-09	Add to Cart and Checkout	User adds gifts to cart and completes checkout.
UC-10	View Gift Details	User views gift description, price, category, and suitability.
UC-11	Inventory Management	Admin manages stock and availability.
UC-12	Manage Offers	Admin adds and updates discounts or promotions.
UC-13	Manage Users	Admin monitors and manages user accounts.
UC-14	View Reports	Admin reviews reports about trends and performance.

Table 8.1: GiftCompass Use Case List

8.3 Grouped Use Cases

The use cases were grouped into four main system processes:

- **Account Management:** Create account, login/logout, and manage profile.
- **Recommendation Process:** Enter recipient details, apply filters, generate recommendations, and view results.
- **Shopping Process:** View gift details, save favorites, add to cart, and checkout.
- **Admin Management:** Manage catalog, inventory, offers, users, analytics, and reports.

8.4 Full Use Case Diagram

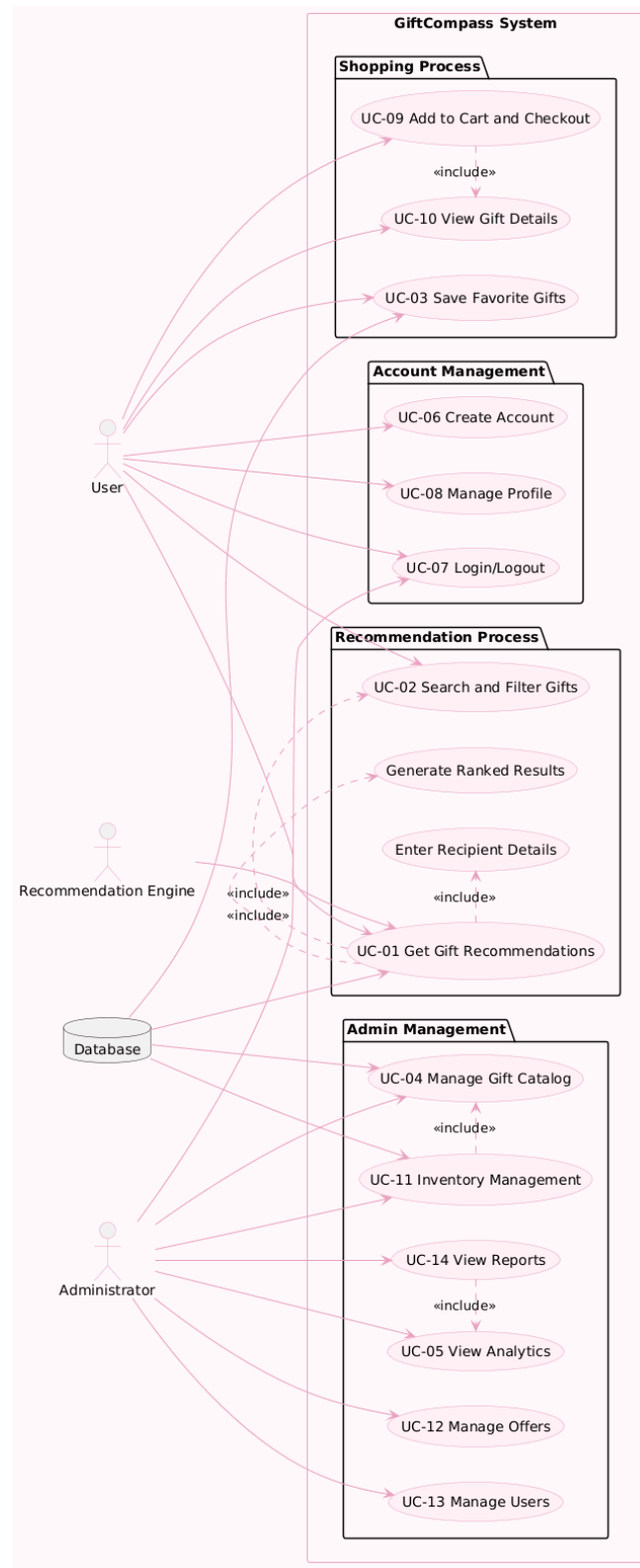


Figure 8.1: Full GiftCompass Use Case Diagram

8.5 Grouped Use Case Diagram

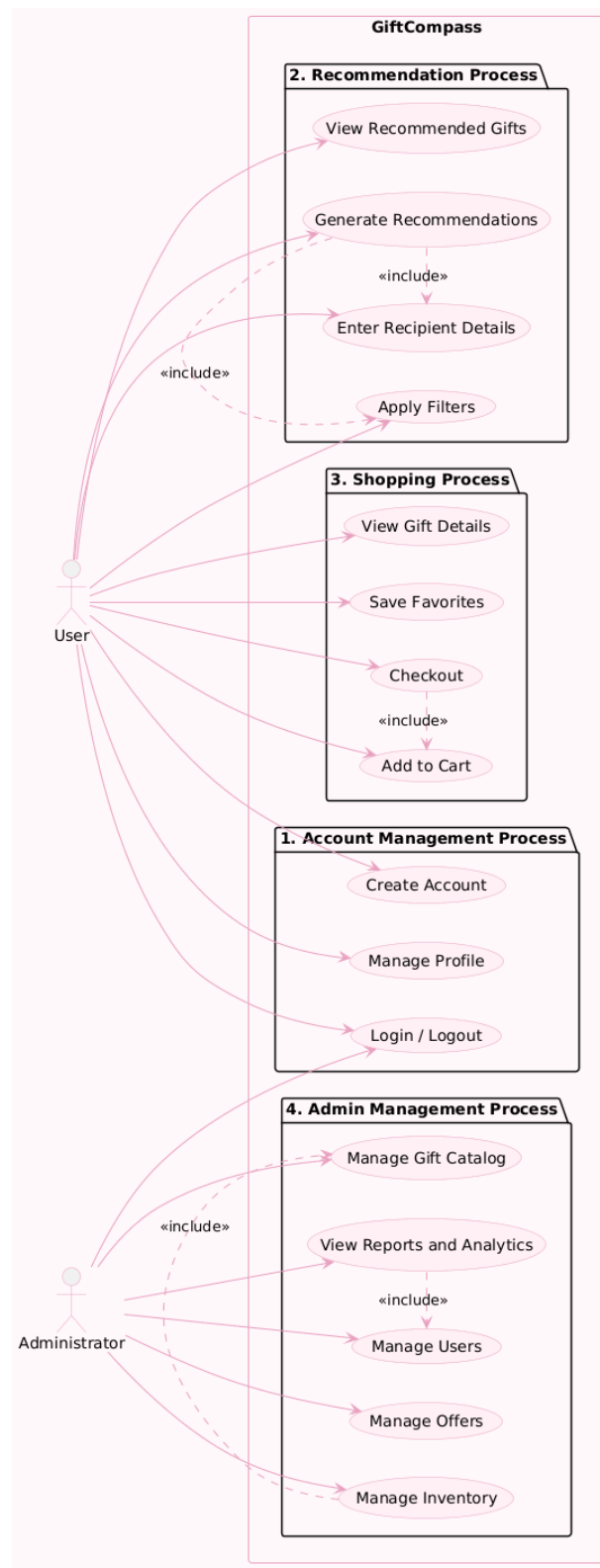


Figure 8.2: Grouped GiftCompass Use Case Diagram

8.6 Use Case Priority

Use Case	Priority	Phase
UC-01: Get Gift Recommendations	Critical	Phase 1
UC-02: Search and Filter Gifts	High	Phase 1
UC-03: Save Favorite Gifts	Medium	Phase 2
UC-04: Manage Gift Catalog	High	Phase 1
UC-05: View Analytics	Medium	Phase 2
UC-06: Create Account	Critical	Phase 1
UC-07: Login/Logout	Critical	Phase 1
UC-08: Manage Profile	Medium	Phase 2
UC-09: Add to Cart and Check-out	High	Phase 1
UC-10: View Gift Details	High	Phase 1
UC-11: Inventory Management	High	Phase 1
UC-12: Manage Offers	Medium	Phase 2
UC-13: Manage Users	Medium	Phase 2
UC-14: View Reports	Medium	Phase 2

Table 8.2: Use Case Priority

Chapter 9

Design Diagrams

9.1 System Context Diagram

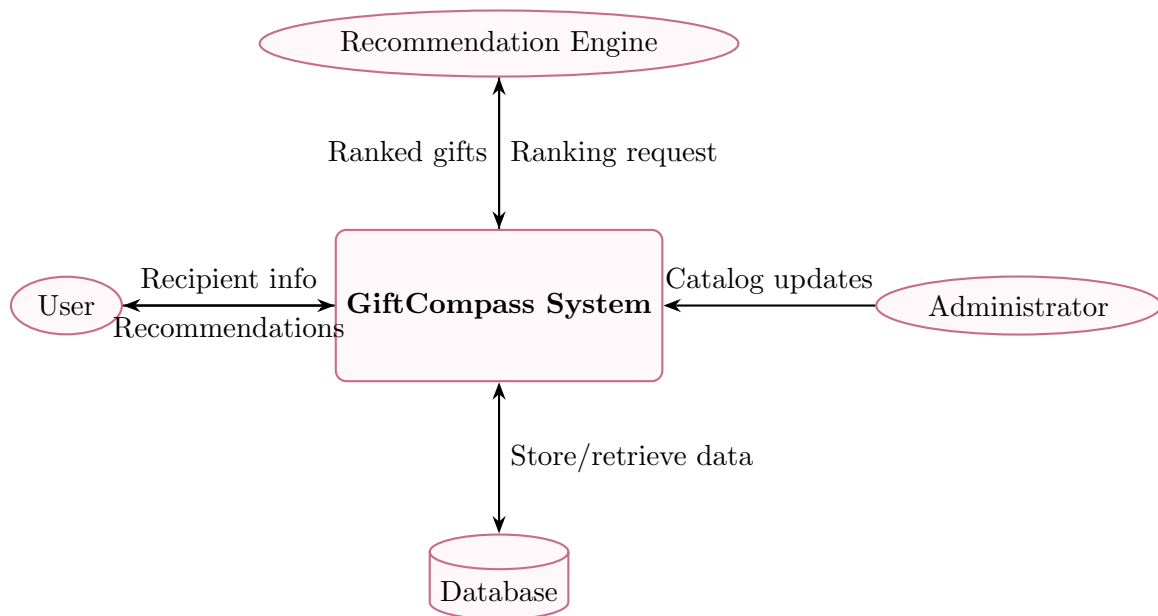


Figure 9.1: System Context Diagram

9.2 Level 1 Data Flow Diagram

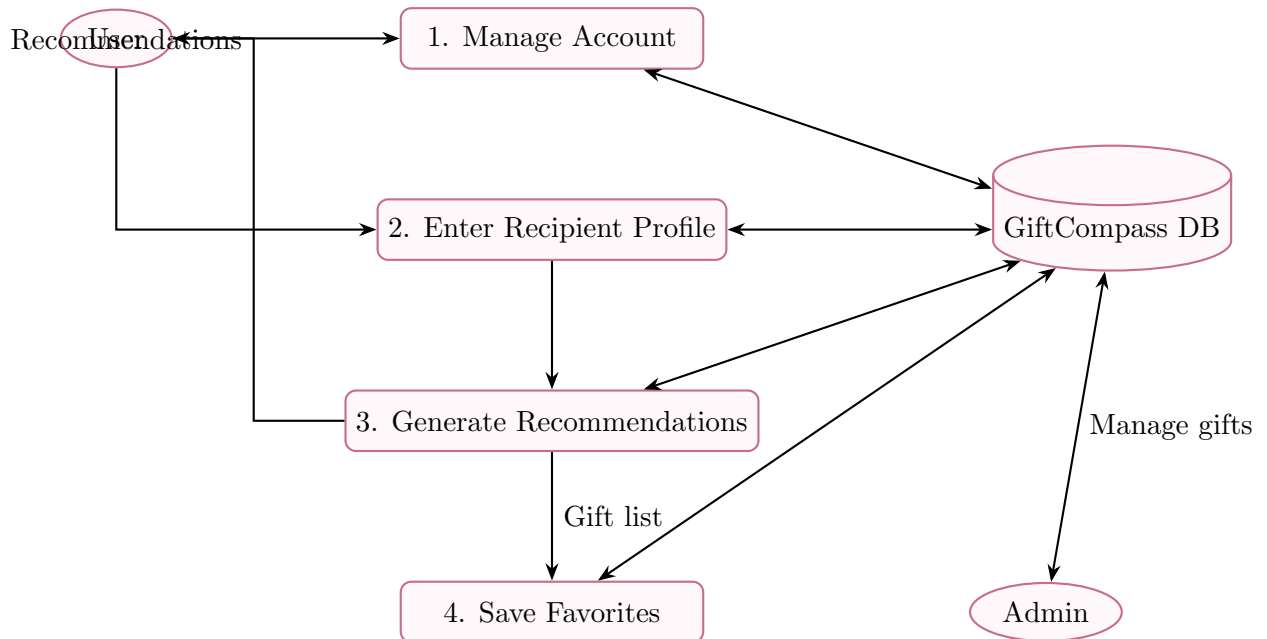


Figure 9.2: Level 1 Diagram for GiftCompass

9.3 Use Case Diagram

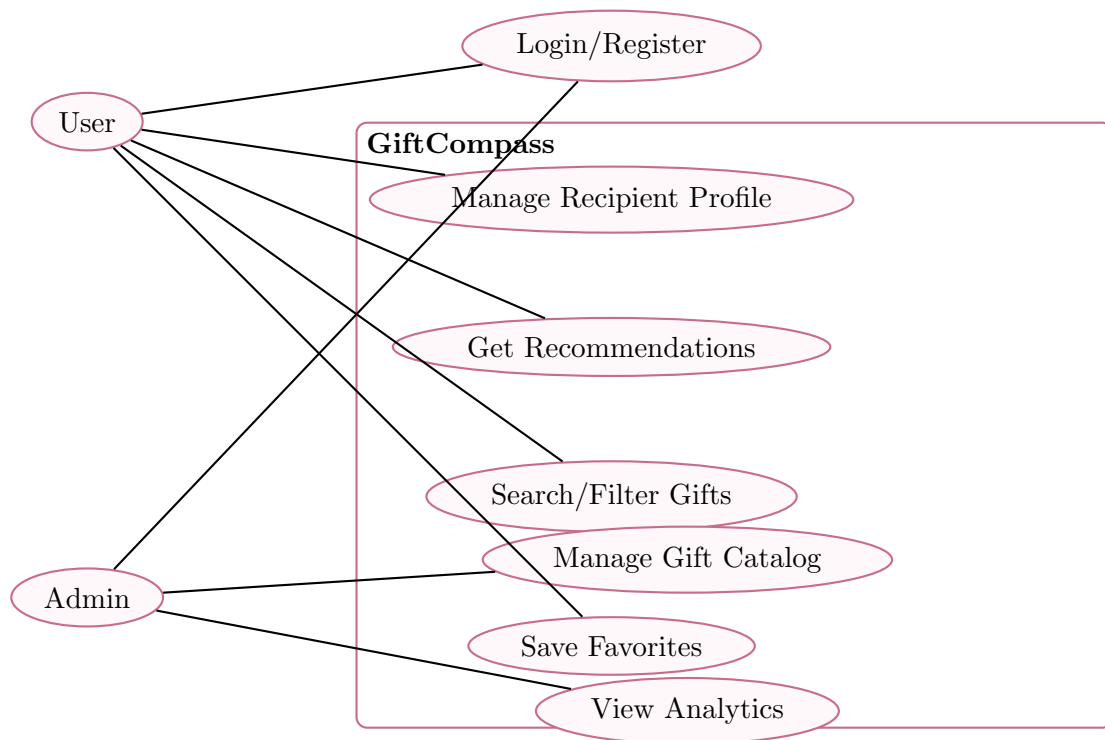


Figure 9.3: Use Case Diagram

9.4 ER Diagram

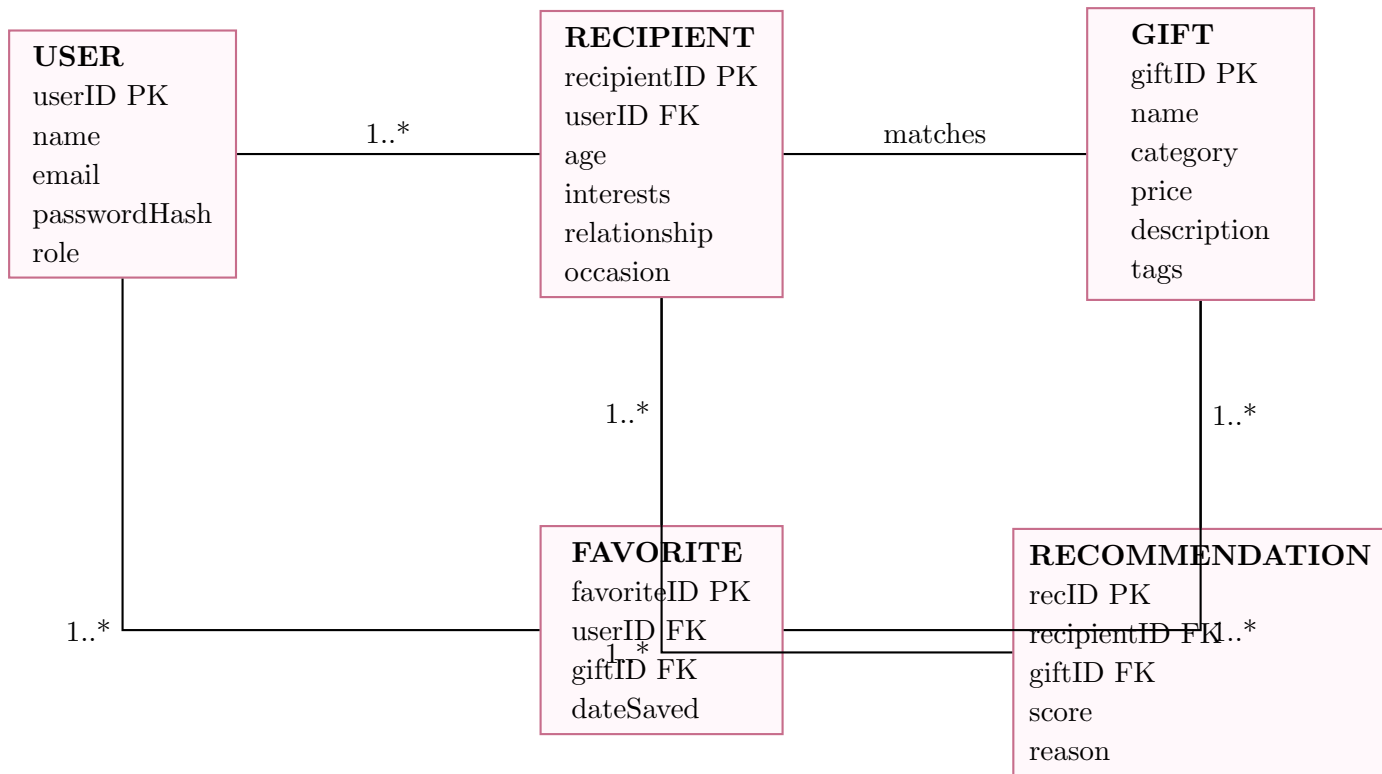


Figure 9.4: Database Entity-Relationship Diagram

9.5 Recommendation Process Diagram



Figure 9.5: Gift Recommendation Process

Chapter 10

Machine Learning and Implementation

10.1 ML Objective

The machine learning objective is to recommend the most suitable gifts based on recipient characteristics and user preferences. The model can use features such as age group, occasion, relationship, interests, budget, gift category, gift price, and previous saved items.

10.2 Dataset Structure

A simple dataset for GiftCompass may contain the following fields:

- Recipient age group
- Relationship
- Occasion
- Interests
- Budget
- Gift category
- Gift price
- Gift rating or popularity score
- Recommendation label or ranking score

10.2.1 Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase was conducted to understand the structure, distribution, and relationships within the GiftCompass dataset. This step is essential for identifying patterns and preparing the data for the recommendation model.

Dataset Overview

The dataset consists of gift items along with their attributes, including product name, category, price, target gender, occasion, interests, and popularity rating. These attributes provide the foundation for generating personalized recommendations.

Data Cleaning

Before performing analysis, several preprocessing steps were applied to ensure data quality:

- Removal of duplicate records
- Handling missing values by either filling or removing incomplete entries
- Standardization of category and interest labels
- Conversion of price values into a consistent numeric format

These steps ensure that the dataset is accurate, consistent, and suitable for analysis.

Descriptive Statistics

Basic statistical analysis was performed to understand the dataset:

- Price range: minimum, maximum, and average values
- Category distribution: number of products in each category
- Occasion frequency: most common occasions
- Interest frequency: most popular interests

Data Visualization and Insights

Several visualizations were used to explore patterns in the dataset:

- **Price Distribution:** A histogram showed that most gifts fall within a mid-range budget, making the system suitable for everyday users.
- **Category Distribution:** A bar chart revealed that categories such as fashion and accessories dominate the dataset, while others require more data.

- **Occasion Analysis:** Birthdays and anniversaries are the most frequent occasions.
- **Interest Analysis:** Common interests include technology, fashion, and hobbies.
- **Price vs Category:** A boxplot indicated that electronics are generally more expensive, while accessories are more affordable.

Relationships Between Variables

Relationships between key variables were analyzed:

- Price and category
- Interests and occasion
- Gender and product type

These relationships help improve recommendation accuracy by identifying meaningful patterns.

Key Findings

The EDA phase revealed several important insights:

- Most users prefer mid-range priced gifts
- Certain categories dominate the dataset and may require balancing
- Birthdays and anniversaries are the most common occasions
- Interests play a major role in personalization

Conclusion of EDA

The EDA process provided a clear understanding of user preferences and product distribution. It also helped identify data gaps and improve dataset quality. These insights support the development of an effective recommendation system based on filtering and ranking techniques.

10.2.2 EDA Python Code with Graphs

Listing 10.1: GiftCompass EDA Python Code

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # -----
5 # GiftCompass sample dataset

```

```

6 # -----
7 data = {
8     "Product": [
9         "Personalized Mug", "Wireless Earbuds", "Skincare Set",
10        "Book Gift Box", "Flower Bouquet", "Smart Watch",
11        "Chocolate Box", "Perfume", "Gaming Headset",
12        "Jewelry Set", "Art Kit", "Fitness Bottle"
13    ],
14    "Category": [
15        "Custom", "Technology", "Beauty", "Books", "Flowers",
16        "Technology", "Food", "Beauty", "Technology",
17        "Accessories", "Art", "Fitness"
18    ],
19    "Occasion": [
20        "Birthday", "Graduation", "Birthday", "Thank You",
21        "Anniversary", "Graduation", "Birthday", "Anniversary",
22        "Birthday", "Wedding", "Birthday", "Graduation"
23    ],
24    "Interest": [
25        "Art", "Music", "Self-care", "Reading", "Decor",
26        "Technology", "Food", "Fashion", "Gaming",
27        "Fashion", "Art", "Fitness"
28    ],
29    "Price": [45, 180, 120, 90, 150, 300, 60, 220, 250, 270, 80,
30              70],
31    "Rating": [4.2, 4.7, 4.5, 4.3, 4.6, 4.8, 4.1, 4.4, 4.7, 4.6,
32               4.2, 4.3]
33 }
34
35 df = pd.DataFrame(data)
36
37 # -----
38 # Data overview
39 # -----
40 print("Dataset Shape:", df.shape)
41 print("\nDataset Info:")
42 print(df.info())
43
44 print("\nMissing Values:")
45 print(df.isnull().sum())
46
47 print("\nDescriptive Statistics:")
48 print(df.describe())
49
50 # -----
51 # Graph 1: Price Distribution
52 # -----

```



```

51 plt.figure(figsize=(8, 5))
52 plt.hist(df["Price"], bins=6, edgecolor="black")
53 plt.title("Gift Price Distribution")
54 plt.xlabel("Price")
55 plt.ylabel("Number of Gifts")
56 plt.tight_layout()
57 plt.show()
58
59 # -----
60 # Graph 2: Category Distribution
61 # -----
62 plt.figure(figsize=(8, 5))
63 df["Category"].value_counts().plot(kind="bar")
64 plt.title("Number of Gifts by Category")
65 plt.xlabel("Category")
66 plt.ylabel("Count")
67 plt.xticks(rotation=45)
68 plt.tight_layout()
69 plt.show()
70
71 # -----
72 # Graph 3: Occasion Frequency
73 # -----
74 plt.figure(figsize=(8, 5))
75 df["Occasion"].value_counts().plot(kind="bar")
76 plt.title("Gift Distribution by Occasion")
77 plt.xlabel("Occasion")
78 plt.ylabel("Count")
79 plt.xticks(rotation=45)
80 plt.tight_layout()
81 plt.show()
82
83 # -----
84 # Graph 4: Average Price by Category
85 # -----
86 plt.figure(figsize=(8, 5))
87 df.groupby("Category")["Price"].mean().sort_values().plot(kind="
    bar")
88 plt.title("Average Gift Price by Category")
89 plt.xlabel("Category")
90 plt.ylabel("Average Price")
91 plt.xticks(rotation=45)
92 plt.tight_layout()
93 plt.show()
94
95 # -----
96 # Graph 5: Price vs Rating

```

```

97 # -----
98 plt.figure(figsize=(8, 5))
99 plt.scatter(df["Price"], df["Rating"])
100 plt.title("Relationship Between Price and Rating")
101 plt.xlabel("Price")
102 plt.ylabel("Rating")
103 plt.tight_layout()
104 plt.show()
105
106 # -----
107 # Simple recommendation scoring
108 # -----
109 def recommend_gifts(interest, occasion, budget):
110     results = df.copy()
111     results["Score"] = 0
112
113     results.loc[results["Interest"].str.lower() == interest.lower
114                 ], "Score" += 3
115     results.loc[results["Occasion"].str.lower() == occasion.lower
116                 ], "Score" += 2
117     results.loc[results["Price"] <= budget, "Score"] += 1
118     results.loc[results["Rating"] >= 4.5, "Score"] += 1
119
120     results = results[results["Price"] <= budget]
121     results = results.sort_values(by=["Score", "Rating"],
122                                   ascending=False)
123
124     return results[["Product", "Category", "Occasion", "Interest",
125                     "Price", "Rating", "Score"]]
126
127 recommendations = recommend_gifts("Technology", "Graduation",
128                                   350)
129 print("\nRecommended Gifts:")
130 print(recommendations)

```

10.2.3 EDA Results

The EDA results show that the GiftCompass dataset contains a variety of gift categories, including technology, beauty, books, accessories, flowers, food, art, and fitness. The price distribution shows that most gifts fall within an affordable mid-range, which supports the system goal of helping users find suitable gifts based on budget.

The category distribution shows that technology and beauty products are among the strongest categories in the dataset. Occasion analysis shows that birthdays, graduations, anniversaries, and weddings are important gift-giving events. The average price by category shows that technology products are usually more expensive, while custom, food, art, and fitness gifts are more affordable.

The price-versus-rating graph helps identify gifts that provide good value. Products with high ratings and reasonable prices are strong candidates for recommendation. These findings support the content-based recommendation method because the system can rank gifts using matching factors such as interest, occasion, budget, category, and rating.

10.3 Recommendation Approach

For the prototype, a content-based recommendation method is suitable. The system compares user input with gift attributes and calculates a score for each gift. Gifts with the highest scores are displayed first. A future version can use collaborative filtering when more user interaction data becomes available.

Listing 10.2: Simple GiftCompass Recommendation Prototype

```
1 import pandas as pd
2
3 # Sample gift catalog
4 gifts = pd.DataFrame([
5     {"name": "Personalized Mug", "category": "custom", "interest":
6         "art", "occasion": "birthday", "price": 45},
7     {"name": "Wireless Earbuds", "category": "technology", "
8         interest": "music", "occasion": "graduation", "price":
9         180},
10    {"name": "Skincare Set", "category": "beauty", "interest": "
11        self-care", "occasion": "birthday", "price": 120},
12    {"name": "Book Gift Box", "category": "books", "interest": "
13        reading", "occasion": "thank you", "price": 90},
14    {"name": "Flower Bouquet", "category": "flowers", "interest":
15        "decor", "occasion": "anniversary", "price": 150}
16 ])
17
18 def recommend_gifts(interest, occasion, budget):
19     results = gifts.copy()
20     results["score"] = 0
21     results.loc[results["interest"] == interest, "score"] += 3
22     results.loc[results["occasion"] == occasion, "score"] += 2
23     results.loc[results["price"] <= budget, "score"] += 1
24     results = results[results["price"] <= budget]
25     return results.sort_values(by="score", ascending=False)
26
27 recommendations = recommend_gifts("music", "graduation", 200)
28 print(recommendations[["name", "category", "price", "score"]])
```

10.4 Stakeholder Dissemination

The model results can be communicated to stakeholders through a simple dashboard. The dashboard should show the number of recommendations generated, most popular gift categories, most selected occasions, and examples of successful personalized recommendations.

Chapter 11

Testing and Validation

11.1 Acceptance Criteria

- A user must be able to create an account with valid information.
- A user must be able to log in within 2 seconds after entering correct credentials.
- The system must generate gift recommendations within 3 seconds.
- Search and filtering must return results that match the selected criteria.
- Admin users must be able to add, update, and delete gifts successfully.
- The system must protect sensitive data using authentication and secure communication.

11.2 Testing Plan

Test Type	Purpose	Expected Result
Unit Testing	Test individual functions such as recommendation scoring and filtering.	Functions return correct outputs.
Integration Testing	Test communication between frontend, backend, and database.	Components work together.
Usability Testing	Check if users can complete the recommendation process easily.	Process completed in 3 steps or less.
Security Testing	Test login, password hashing, and authorization.	Unauthorized access is blocked.
Performance Testing	Measure response time under multiple requests.	95% of requests respond within 2 seconds.

Table 11.1: Testing Plan

Chapter 12

UI/UX Design

12.1 Main Screens

12.2 Application Screens

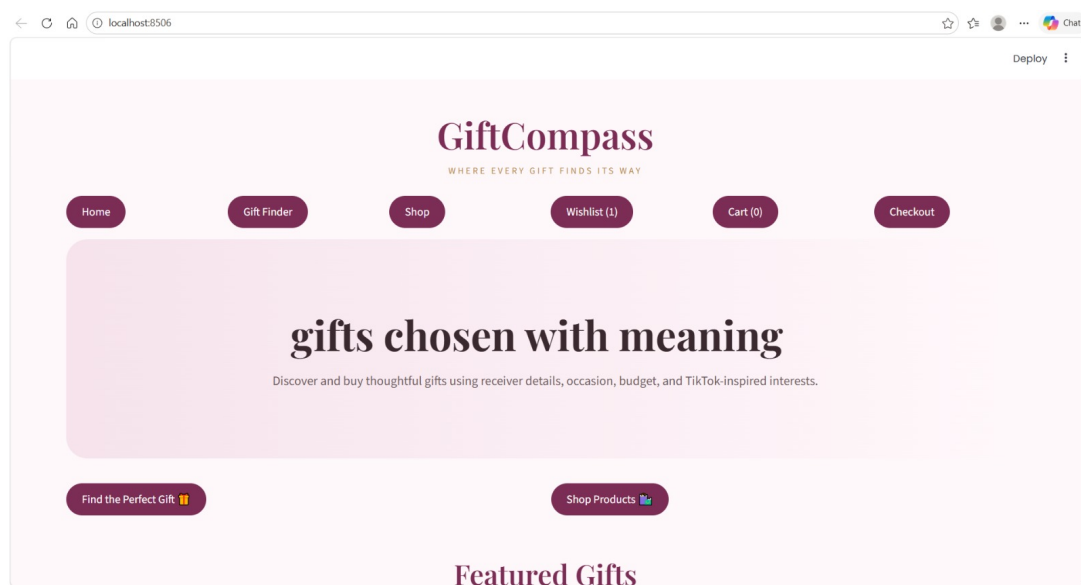


Figure 12.1: Main Page

The screenshot shows the GiftCompass website interface. At the top, there's a navigation bar with links: Home, Gift Finder, Shop, Wishlist (0), Cart (0), and Checkout. Below this is a section titled "Find the Perfect Gift". The form includes several input fields: "Who is this person related to you?" (dropdown menu with "Friend" selected), "Age" (input field with "20"), "TikTok username" (input field with "@receiver"), "Occasion" (dropdown menu with "Birthday" selected), "Budget in SAR" (slider set to 500), and "Extra interests" (dropdown menu with "Choose options" selected). A button labeled "Analyze TikTok Reposts & Recommend Gifts" is at the bottom of the form. The footer text reads: "GiftCompass © 2026 | Smart TikTok-Based Gift Recommendation & Shopping Platform".

Figure 12.2: Recommendation Form

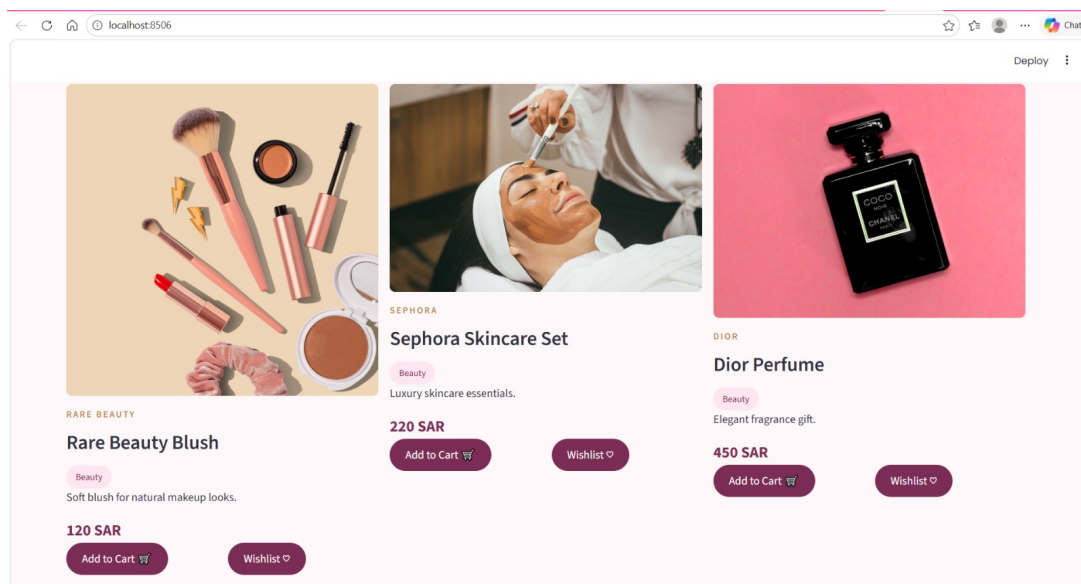


Figure 12.3: Gift Suggestions Page

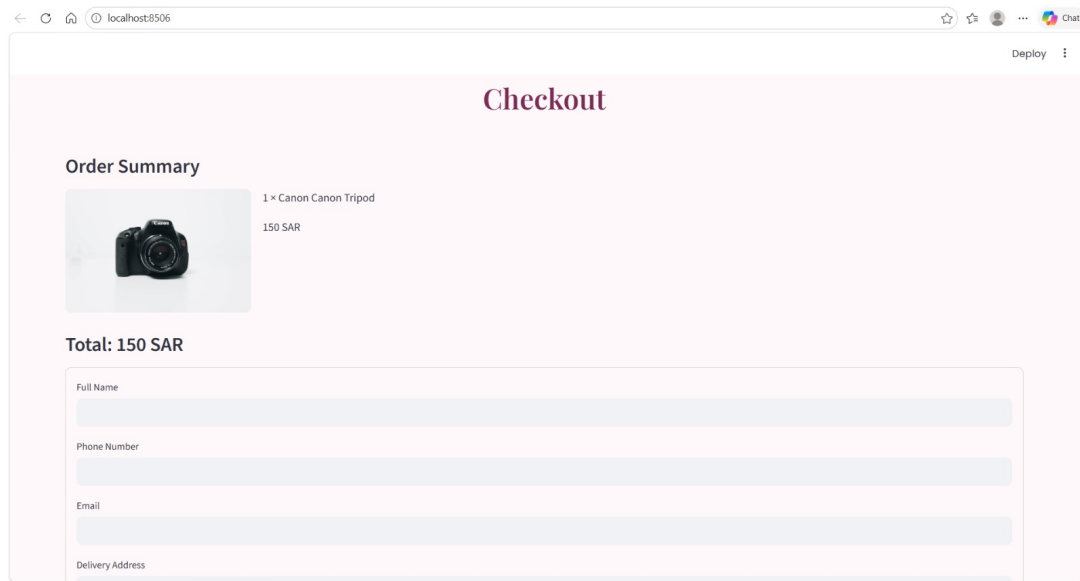


Figure 12.4: Checkout Page

The GiftCompass user interface should be simple and friendly. The main screens include:

- Login and registration page.
- Recipient information form.
- Gift recommendation results page.
- Gift details page.
- Favorites page.
- Admin catalog management page.
- Analytics dashboard.

12.3 System Flow Diagram

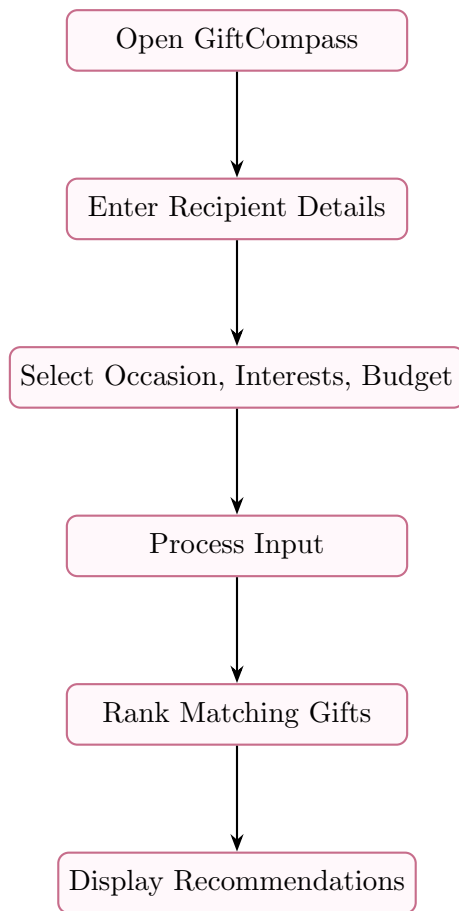


Figure 12.5: GiftCompass System Flow Diagram

Chapter 13

Discussion and Results

GiftCompass successfully demonstrates how a recommendation-based system can improve the gift selection process. The project results show that collecting structured recipient information such as age, relationship, occasion, interests, and budget helps the system narrow down gift options and present more relevant suggestions.

The prototype interface shows that users can move from the main page to the recommendation form, view suggested gifts, add items to the cart, and proceed to checkout. This supports the main project objective of reducing the time and effort needed to find suitable gifts.

The recommendation approach used in the project is content-based. It compares the user input with gift attributes and ranks the gifts according to matching factors. This approach is suitable for the current prototype because it does not require large amounts of user history data. In future versions, the system can be improved using collaborative filtering, user feedback, and real purchase behavior.

Overall, the project outcomes show that GiftCompass is feasible, useful, and scalable. The system provides a clear solution to the gift-selection problem and can be expanded into a stronger AI-based recommendation platform.

Chapter 14

Conclusion

GiftCompass provides a practical solution for users who struggle to choose thoughtful and suitable gifts. The system uses structured recipient information and recommendation logic to generate personalized suggestions. The project demonstrates clear software engineering planning through feasibility analysis, market comparison, requirements specification, design patterns, use cases, system diagrams, and a prototype recommendation approach.

The expected outcome is a user-friendly web platform that saves time, improves confidence in gift selection, and creates a more organized gift discovery experience. Future work can include real e-commerce API integration, more advanced machine learning models, user feedback learning, Arabic language support, and partnerships with local gift shops.

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Appendix A

Glossary

- **GiftCompass:** A system that recommends gifts based on recipient information.
- **User:** A person who uses the system to search for gifts.
- **Admin:** A system manager who maintains gift data.
- **Recipient Profile:** Information about the gift receiver.
- **Recommendation Engine:** The component that generates gift suggestions.

Appendix B

Business Model Canvas Summary

Canvas Element	GiftCompass Summary
Value Proposition	Personalized, fast, and thoughtful gift recommendations.
Customer Segments	Students, professionals, family members, friends, and gift shoppers.
Channels	Web platform, social media, and future mobile app.
Customer Relationships	Self-service, guided forms, and personalized suggestions.
Revenue Streams	Future affiliate links, promoted gifts, premium features, and seller partnerships.
Key Resources	Gift database, recommendation logic, web platform, and user data.
Key Activities	Catalog management, recommendation generation, user support, and platform updates.
Key Partners	Local gift shops, e-commerce platforms, and delivery services.
Cost Structure	Development, hosting, maintenance, marketing, and API integration.

Table B.1: Business Model Canvas Summary

The Buisness Model Canvas

Designed for:

Dr Passent

Designed by:

Sedra, Kamar, Layan

Date:

28-1-2026

Version:

01



Figure B.1: GiftCompass Business Model Canvas Visualization